PROBABILITY AND RANDOM VARIABLES

Omkar Hambire(16)
Mehul Hingu(17)
Sanchit gaikwad (11)
Aaditya ghanekar(12)

What is Probability ?

- Probability is the study of chance and uncertainty.
- It helps us measure how likely an event is to occur.
- Distributions show how probabilities are spread across different outcomes.
- Used in science, engineering, business, games, and daily life.

Basic Concepts of Probability

Experiment: Any process with uncertain outcome (e.g., tossing a coin).

Sample Space (S): Set of all possible outcomes.

Event (E): A subset of sample space.

Probability Formula:

p(E)=(Total number of possible outcomes)/(Number of favorable outcomes)

Example: Probability of drawing a red card from deck = 26/52 = 0.5

Poisson Distribution :

Definition :

A discrete random variable X follows a Poisson distribution with an average rate of λ (lambda) if it represents the number of events occurring in a fixed interval. It is denoted as $X \sim \text{Pois}(\lambda)$.

Formulas: Probability Mass Function (PMF):

$$P(X = k) = (\lambda^k * e^{(-\lambda)}) / k!$$

- $P(X = k)$ is the probability of observing exactly k events.
- λ is the average/expected number of events in the interval.
- $k!$ is the factorial of k (e.g., $5! = 5\times4\times3\times2\times1 = 120$

.E is Euler's number (approximately 2.71828).

🎲 Number Guessing Game

Title: Number Guessing Game

Points:

- A simple interactive game that demonstrates randomness and probability.

How it works:

- The computer randomly selects a number between 1 and 100.
- The player tries to guess the number.
- After each guess, the game gives hints: “Higher” or “Lower.”

Goal: Find the number in as few guesses as possible.

🎲 Random Variable in the Game

In this game:

```javascript
function rand(){ return Math.floor(Math.random()*100)+1 }
```

- Math.random() generates a continuous random variable between 0 and 1.
- Multiplying by 100 and flooring it gives a discrete random variable taking integer values from 1 to 100.
- This is a discrete uniform random variable because each number from 1 to 100 has equal probability.

🎯 Probability of Guessing

Let's say the player guesses randomly without using logic:

- Probability of success in 1 attempt:

$$P(\text{correct in 1st try}) = \frac{1}{100} = 0.01$$

- Probability of success in k-th attempt (random guessing without memory):

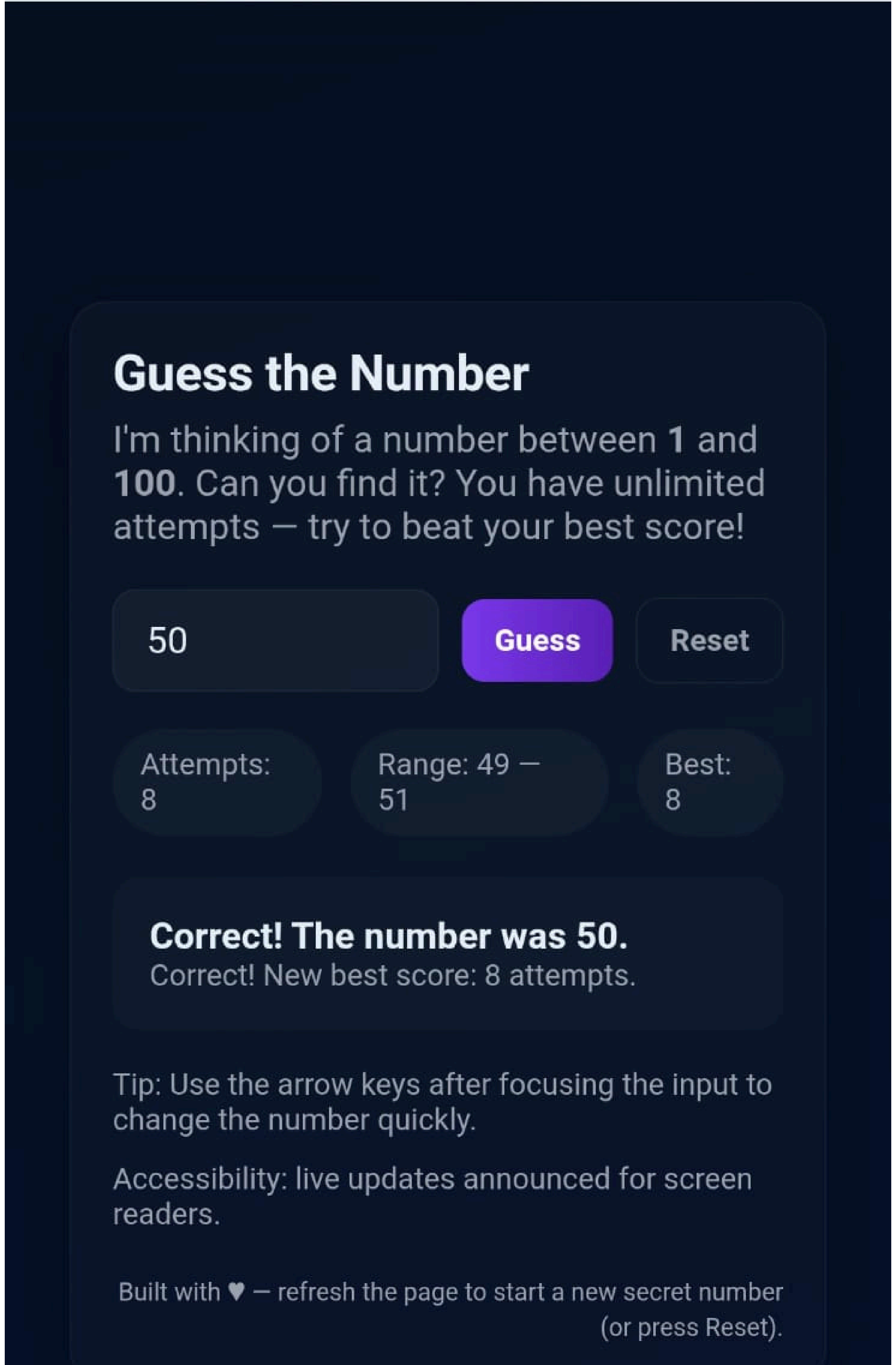
Still $\frac{1}{100}$ each time, since each guess is independent.

- With memory (not repeating numbers):

The probability of success on the k-th guess:

$$P(\text{success at k}) = \frac{1}{101 - k}, \quad k = 1, 2, \ldots, 100$$

GAME INTERFACE



Real Life Applications

1. Weather Forecasting

- Probability predicts chances of rain, storm, or sunshine.
- Random variables model uncertain weather conditions.

2. Traffic Management

- Predicting traffic flow at signals.
- Random variables represent number of vehicles passing per minute.

3. Games And Gambling

- Probability predicts winning chances in cards, dice, lottery, casino games.

4. Stock market and Finance

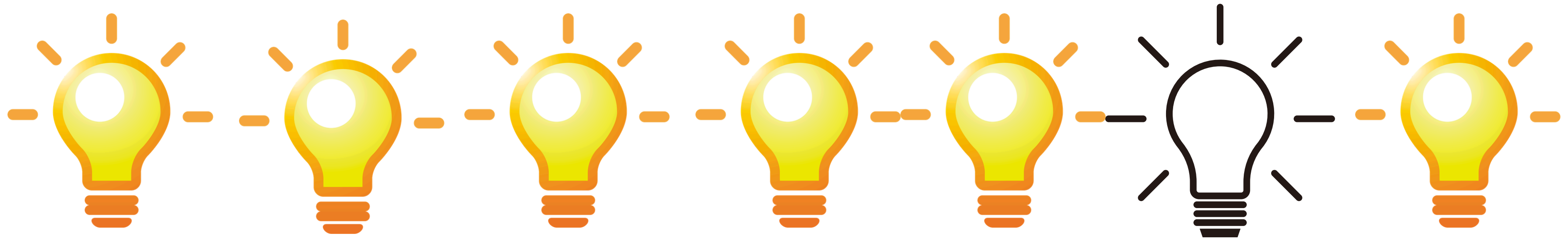
- Random variables model stock price fluctuations.
- Probability helps in risk management and investment decisions.

5. Quality Control in Industries

Probability distributions (like normal distribution) used to check defect rates in products.

6. Communication Systems

Engineers use probability to model noise and errors in signals.



Conclusion

Probability and random variables provide powerful tools to model uncertainty.

Add a little bit of body text

They are essential for solving real-world engineering problems.

Understanding distributions helps in decision making under uncertainty.

Thank you